

Ahmed Hussein Ali Abdelsalam Hussein

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Professional Summary

Data Science graduate from the Faculty of Artificial Intelligence, Menofia University, with a GPA of 3.67 and hands-on experience in machine learning, RAG, LLMs, NLP, computer vision, OCR, digital forensics, data analytics, and cybersecurity AI. Strong in Python, SQL, model evaluation, dashboarding, technical communication, and applied research, with experience building end-to-end AI and analytics solutions. Interested in developing reliable data-driven products, contributing to applied research, and sharing technical knowledge in collaborative professional or academic environments.

Education

Faculty of Artificial Intelligence, Menofia University

2022 - 2026

Bachelor of Science in Artificial Intelligence - Data Science Major | GPA: 3.67

Menofia, Egypt

Overall Grade: Excellent with Honors | **Academic Rank:** 3rd in the Data Science Department (Years 2, 3 & 4)

Academic Profile

Relevant Coursework: Data Science, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Analysis, Business Intelligence, Financial Analytics, Cybersecurity Analytics, OCR and Document AI.

Research Interests: Retrieval-Augmented Generation, Large Language Models, Explainable and Trustworthy AI, Computer Vision, Applied Machine Learning, Data-Centric AI, and AI for Cybersecurity.

Professional & Teaching Strengths: Python and SQL support, machine-learning demonstrations, data-analysis tutorials, dashboard development, technical presentations, model evaluation, applied research, team collaboration, and student project guidance.

Experience

AI & Data Research Intern - Compumacy | 2 Months

- Contributing to ArabicPsychBench, an Arabic psychiatry benchmark research project designed to evaluate Large Language Models (LLMs) and support preparation of a research paper.
- Curated and structured content from 16 Arabic psychiatry and clinical psychology books, organizing 535 Q&A items, 217 clinical case candidates, MCQs, and dialogue-based data while preserving source and page-level provenance.
- Built Python/OCR preprocessing and task-planning pipelines to map source items to Mental QA, Diagnosis, Treatment, Clinical Approach, Classification, Treatment Follow-Up, and Sequential QA tasks.
- Implemented quality gates, source-fidelity checks, target-leakage detection, review queues, and validation outputs to prepare the dataset for expert review and subsequent LLM evaluation experiments.

Projects

AEGIS Graduation Project - Android Security Platform | Python, FastAPI, React, Docker, Ollama, RAG, MITRE ATT&CK | [Repo](#)

- Led the APK Studio subsystem and implemented APK intake, static and dynamic analysis, emulator-based testing, evidence lineage, review workflows, and evaluation pages.
- Integrated local AI assessment, risk scoring, MITRE ATT&CK mapping, traceable evidence references, and PDF-ready reporting for analyst-driven mobile security review.

DeepTrace AI - Digital Forensics Platform | Python, FastAPI, Streamlit, OpenCV, Tesseract, PyMuPDF, SQLAlchemy, Ollama | [Repo](#)

- Built a local-first forensic workstation for image and PDF evidence analysis with Arabic-English OCR, metadata and ELA inspection, SHA-256 integrity checks, and exact/near-duplicate detection.
- Separated deterministic analysis from optional local LLM summaries with bounded timeouts, explainable risk signals, secure evidence ingestion, performance instrumentation, persistence, and automated tests.

LangGraph Multi-Agent Mobile Security Assistant | Python, LangGraph, RAG, FAISS, Qdrant, Ollama, MASVS, LangSmith | [Repo](#)

- Built a guarded multi-agent architecture with routing, memory, Qdrant retrieval, MASVS knowledge, optional web search, and grounded bilingual response generation.
- Added FAISS long-term memory, prompt-injection safeguards, LangSmith observability, and a professional Streamlit interface.

Document Q&A System (RAG) | Python, Streamlit, FAISS, LangChain, Sentence-Transformers, Gemini API | [Repo](#)

- Implemented PDF extraction, chunking, embeddings, semantic retrieval, and context-grounded answer generation in an interactive document-analysis application.

Eco-Guard: Plant Disease AI | Python, TensorFlow, Keras, OpenCV, Plotly | [Repo](#)

- Designed a CNN pipeline for plant-disease detection and severity estimation, with heatmaps, analytics, treatment recommendations, and report export.

AI-Powered Financial Advisory Platform | Python, Streamlit, LangChain, SQLite, Pandas | [Repo](#)

- Developed a bilingual financial analytics platform with health scoring, savings and debt recommendations, sensitivity analysis, interactive charts, and history tracking.

OllamaMed - Medical Report Summarizer | Python, Streamlit, Ollama, Llama 3, Tesseract OCR, OpenCV | [Repo](#)

- Built a privacy-focused offline pipeline that extracts text from medical PDFs and images, summarizes reports locally, highlights key findings, and explains technical terminology.

Home Energy Tracker - Time-Series Forecasting | Python, Streamlit, Pandas, Statsmodels/SARIMAX | [Repo](#)

- Built ETL/EDA workflows and an interactive dashboard for daily, monthly, seasonal, and sub-metered household energy-consumption analysis.
- Integrated weather variables with SARIMAX forecasting to support short-horizon consumption predictions and contextual usage insights.

Projects (Continued)

Interactive Plotly Dashboard Portfolio | Python, Dash, Plotly, Pandas | [Repo](#)

- Built four interactive analytics dashboards covering NYC Airbnb, building permits, COVID-19 business impact, and world happiness.
- Implemented KPI cards, filters, maps, trend analysis, aggregations, responsive layouts, and downloadable filtered data for exploratory analysis.

AdventureWorks Excel & SQL Analysis | SQL Server, Excel, Power Query | [Repo](#)

- Collaborated on SQL preprocessing, table joins, calendar logic, and interactive Excel dashboards for sales, product, territory, logistics, and employee analysis.
- Applied KPI validation and data-quality checks to surface inconsistencies and support multi-year business insights from AdventureWorks data.

AdventureWorks Power BI Analytics | Power BI, DAX, SQL, Excel | [Repo](#)

- Developed an interactive BI dashboard for sales, products, customers, regions, and operational KPIs with multi-dimensional filtering and trend analysis.
- Added geographic comparisons, top-product rankings, key-influencer analysis, and drill-down views for decision-oriented reporting.

Adidas Sales & Profit Analysis Dashboard | Tableau, CSV, Data Visualization | [Repo](#)

- Built an interactive Tableau dashboard for U.S. Adidas sales covering revenue, profit, units sold, retailer and product performance, regions, and sales channels.
- Used geographic maps, KPI tracking, monthly trends, profitability comparisons, and channel analysis to communicate business insights.

Amazon Reviews Sentiment Analysis | Python, TensorFlow, Pandas, NLTK, Streamlit | [Repo](#)

- Performed EDA and text preprocessing, trained an LSTM-based sentiment classifier, evaluated it with classification metrics and confusion matrices, and built a Streamlit interface.

Diabetes Analysis Project | Python, Pandas, NumPy, Matplotlib | [Repo](#)

- Performed exploratory data analysis, visualization, and correlation analysis on a diabetes dataset to communicate healthcare-oriented patterns and insights.

Additional Selected Work: [Threat Hunting Assistant](#) (RAG and MITRE ATT&CK); [YouTube Video Summarizer](#) (BART, PyTorch, Streamlit)

Professional Training

- ITI - Business Intelligence Development; NTI - Data Analysis Skills for the Freelance Professional.
- LLM Internship at Tips Hindawy; Data Science Internship at Cognorise Infotech; Machine Learning Internship at Prodigy Infotech.
- Eyooth Data Analytics Bootcamp - applied data analysis, visualization, machine learning, and practical analytics workflows.

Technical Skills

Programming: Python, SQL

AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision, RAG, LLMs, OCR, Sentiment Analysis, Model Evaluation, Time-Series Forecasting

Data Analytics & BI: Pandas, NumPy, Matplotlib, Plotly, Excel, Power BI, Tableau, Power Query, DAX, SQL Server, EDA, Data Cleaning, Dashboarding

Frameworks/Libraries: TensorFlow, Keras, PyTorch, scikit-learn, FastAPI, LangChain, Hugging Face Transformers, FAISS, Sentence-Transformers, OpenCV, Streamlit

Tools/Data: Gemini API, Ollama, Tesseract OCR, PyMuPDF, ChromaDB, Qdrant, SQLite, SQLAlchemy, Git/GitHub, Docker, Ngrok

Cybersecurity: Digital Forensics, MITRE ATT&CK, MASVS, Androguard, Scapy

Professional Strengths: Applied research, technical writing and presentation, communication, problem solving, collaboration, adaptability, and continuous learning.